

so i am building a harambee website and am stuck at payment integration where the client wants to use the paybill they use during tithe where congregation use paybil and accounts number as there name but here its intendedt for the user input his numbers and a push message to be sent to him to enter thempesa pin,,,like the one in supermarkets but i know both are 2 different systems so i was thinking on how one will be in backendn to execute very fast and the other one to operate in the front both linked up

Architected dual-flow payment diagram merging C2B and STK paths >

Good news: you're not actually dealing with two separate payment systems that need bridging — you're dealing with two different ways of triggering the same M-Pesa Paybill, and Safaricom's Daraja API has a distinct integration pattern for each. Once you see them as two API flows hitting one shared backend, the "linking up" part becomes straightforward.

The tithe-style flow (manual paybill + account number) is what Safaricom calls **C2B (Customer to Business)**. The customer initiates everything from their own phone — they open M-Pesa, select Pay Bill, type your org's paybill number, type whatever they want in the account field (their name, in your case), enter the amount, and confirm with their PIN. Your role here is passive: you register a Validation URL and a Confirmation URL with Safaricom ahead of time, and when someone completes a payment, Safaricom calls your Confirmation URL with the transaction details (amount, M-Pesa receipt number, phone number, and whatever string they typed into the account field). There's no "push" involved and no initiation from your side at all — you're just listening for a webhook.

The supermarket-style flow is **STK Push** (Lipa na M-Pesa Online / M-Pesa Express). Here your backend is the initiator: the user types their phone number and amount into your website, your server calls Safaricom's STK Push endpoint with the paybill shortcode, an account reference, and a callback URL, and Safaricom sends the prompt to the user's phone for them to enter their PIN. Once they do (or it times out, ~60–90 seconds), Safaricom hits your callback URL with the result, and you update your records.

The thing that makes these "link up" naturally is that both can run against the *same* paybill shortcode, provided it's registered for both C2B and STK Push with Safaricom (this is standard for an organization paybill — it's not an either/or registration). So your harambee site doesn't need two payment systems, it needs one backend service exposing two endpoints: one that's reachable by Safaricom for C2B confirmations (passive listener), and one that actively calls Safaricom for STK push and then waits on its own callback. Both write into the same contributions table, just tagged with a different payment_method so you know which path each record came from.

A couple of things worth planning for early since they trip people up: the C2B account-number field is free text the donor types themselves, so matching "JOHN K" or a misspelled name to an actual pledge record needs either a reconciliation UI for your admin, or you ask people to use a unique code instead of their name. Also, your callback URLs need to be public HTTPS endpoints before Safaricom will accept registration, and you'll want idempotent handling on both callbacks since Safaricom can retry delivery.

The blue path is the tithe-style C2B flow, teal is the STK push flow, and they both funnel into the same gray backend layer — that's the "linking up" part, it isn't two separate backends, just two listeners on one service.

To get more specific with you (code, framework choices, etc.), it'd help to know what you're building this on. Are you using a particular backend stack already (Flask/Django/Node/etc.), and do you already have Daraja API credentials (consumer key/secret, and is the paybill already set up for both C2B and Lipa na M-Pesa Online, or do you need to request that from Safaricom/your sacco/bank first)?